Venkata Lakshmi Parimala Pasupuleti

Senior AI Engineer · Agentic Workflows · LangChain · MCP · Multi-Agent · LLM Evaluation · Guardrails · Observability · Python · AWS

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SUMMARY

Senior AI engineer with 9+ years building production agentic systems, LLM-powered workflows, and evaluation infrastructure end-to-end. Built **Leasel** - a production agentic workflow platform using LangGraph (LangChain ecosystem) with tool calling, durable state, MCP server integration, hybrid search RAG, HITL guardrails, and a CI evaluation harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source LLM evaluation platform with offline evals, LLM-as-judge scoring, observability, failure handling, and cost-aware multi-provider routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama. Currently at Intuit integrating Anthropic Claude into production financial services: autonomous workflows, tool integrations, structured outputs, guardrails, and evaluation infrastructure. At Entrata (7+ years, 100K+ users), owned mission-critical data pipelines and async distributed systems in a regulated multi-tenant SaaS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Agentic Workflows · Autonomous Task Execution · Orchestration | LangChain · LangGraph · Agent Orchestration Frameworks

MCP · Model Context Protocol · Tool Integrations | Multi-Agent Systems · Tool Use · Durable State

LLM Evaluation · LLM-as-Judge · Golden Datasets · CI Regression | Guardrails · Failure Handling · Safety Controls · HITL

Observability · Monitoring · Tracing · OpenTelemetry | RAG Pipelines · Hybrid Search · Embeddings · Retrieval

Third-Party Platform Integrations · API Design · Webhooks | Python · FastAPI · TypeScript · Node.js

AWS (Lambda · Bedrock · SQS · API Gateway) · Docker · Kubernetes | Distributed Systems · Async Processing · Backend Engineering

End-to-End Ownership · Senior IC · Remote-First Collaboration

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Autonomous AI workflows:** Design and build LangChain-based agentic workflows integrating Anthropic Claude into production financial services - tool calling, retrieval-grounded context management, structured outputs, and third-party platform integrations; guardrails with input validation, sensitive-data handling, and fallbacks.
- **Evaluation and observability:** Build LLM-as-judge evaluation harnesses with golden datasets and CI regression gates; monitor production AI workflows via structured logging and CloudWatch; perform qualitative error analysis and turn failures into eval coverage. Document patterns and playbooks for team reuse.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical data pipelines and async distributed systems end-to-end - HUD compliance ETL (180K records, zero violations across 6 years), RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Agentic Workflow Platform: LangGraph · MCP · RAG · Guardrails · 2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production agentic workflow platform: **8-agent LangGraph orchestration** with tool/function calling, durable state, conditional routing; **MCP server integration** for shared tool access; **hybrid search RAG** (OCR ingestion, chunking, TF-IDF + semantic reranking, citation grounding); **guardrails** enforcing HITL for regulated decisions; **CI eval harness** (LLM-as-judge, golden datasets: 93.6% precision, 1,240 test cases). Three consuming product surfaces share one data and retrieval layer.

AgentProbe - LLM Evaluation & Observability Platform 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source eval and observability platform: offline golden-dataset evals, LLM-as-judge scoring, online quality metrics, CI regression gating, failure handling with structured error classification; cost-aware routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full tracing and structured logging per run.

AI on FHIR — Clinical NLP & Enterprise API Integration 2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Third-party platform integration, OAuth-secured APIs, HIPAA-style RBAC, Docker-containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017